SAMPLE

Building the Military AI of Tomorrow

The Decision That Cannot Be Delegated

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About this Series

This free sample contains the Introduction and Chapter 1. The complete book contains the full contents above.

Introduction

Military artificial intelligence is what happens when the calculation inside the weapon stops being an engineering artifact fixed on a drafting table and becomes a statistical artifact grown from a training set. Every weapon in history has made decisions; the sea mine decided when a hull touched it, the heat-seeking missile decided when a plume crossed its retina, and the guidance computer decided, thousands of times a second, how to trim its fins. What those systems shared was a decision procedure a person could write down, inspect, and audit line by line, because a person had written every line.

The machine that learns breaks that inheritance. Its decision procedure is not written; it is induced, from examples, by an optimization process whose output no engineer can fully read, whose behavior on cases it has not seen cannot be fully predicted, and whose failures arrive not as faults in a logic but as properties of a distribution. The field this book describes exists because that single change, from the authored rule to the grown rule, rewrites every contract a military had with its machines: how it tests them, how it trusts them, how it commands them, and above all who answers when they are wrong.

The first lesson of the subject is that the definition is a battlefield. A missile's autopilot and a drone's lost-link orbit are automated in the strict sense: fixed procedures, executed without a hand on them, approved by a person before the mission and bounded by the geometry the person drew. Artificial intelligence, in the sense this field must police, is the class of systems whose behavior was not specified but trained, whose rules were not written but inferred, whose edge cases were not enumerated but statistically absorbed. The two classes share hardware, share vocabulary, and share vendors, and the public conversation blurs them constantly, sometimes innocently and sometimes for advantage, because a manufacturer who can call his autopilot "AI" buys a budget line and a headline.

This book draws the line sharply and defends it throughout: the audit question, whether a system's decision procedure can be traced to decisions a person can be named for, is the line that matters, and the chapters will apply it to every claim that walks past wearing the label.

The second lesson is that the machine is honest in a way its procurement is not. A classifier trained on ranges does not care about doctrine; it reports the world as its training set taught it to see the world, and if the training set was built from desert imagery, then the machine in the forest is not a sensor suite but a rumor. This honesty is the field's central asset and its recurring scandal,

because every failure the machines produce in the field is, at bottom, a disclosed fact about the data someone collected, and every such disclosure names an institution rather than a glitch. The chapters that follow treat the training set as what it structurally is: the system's true design document, written not by the engineers but by whoever chose the examples, and therefore the artifact a fielded system's behavior must always be audited against.

The third lesson, and the one the decade's arguments will be fought over, is that autonomy is a delegation of judgment, not a delegation of accountability. A commander can hand a machine the authority to act within a box he drew; he cannot hand it the responsibility for what it does inside the box, because responsibility is not a parameter that can be passed across an interface, it is a social fact that requires a being that can be asked questions under oath. The record of the field's first century of automation, and of its first decades of learning systems, is a record of institutions behaving as if this transfer were possible: signature blocks attached to machine outputs, review boards convened to bless statistical procedures, doctrines written on the assumption that a documented delegation is a discharged duty. The book disagrees, in the first person, and spends a chapter on the argument.

This volume follows the series method. It defines the subject from its anatomy outward, works the governing arithmetic in plain numbers, prices every claim against the field's own record, and refuses every statistic a budget cycle can outlive. No manufacturer is named, no platform is endorsed, and no engagement is presented as history: the episodes are composites drawn from public reporting and named as such, because the series' standard is that a claim must survive the death of the product that inspired it. The chapters build the structure in the order the structure requires: the definition and its impostors first, because a field that cannot classify its machines cannot govern them; then the learning machine itself, its training data and its arithmetic; then perception, the sensing pipeline where most of the field's money and most of its failures live; then the decision machinery, the levels of autonomy and the boxes a person draws; then the human in and around the loop, where the signatures and the responsibility live; then the adversary, who gets a vote, and the machine's brittle genius under attack; then the collective, because networked machines have their own arithmetic of consensus and its own failure shapes; then verification, the discipline that decides whether any of it can be fielded; then the record, where the claims are priced against what the institutions actually did; and finally the decade ahead, stated in honest probabilities.

The reader this book is written for holds any of three jobs. A builder needs the

mechanics: what a training set actually costs to collect, what a confidence score actually promises, what a confusion matrix actually implies for a weapon. An operator or analyst needs the doctrine: what the machine is good at, what it is quietly terrible at, and what its literature is reluctant to admit. A policymaker needs the consequences: where the delegation is legally thin, where it is politically explosive, and which decisions must be structured so that a person can be named for them in a hearing room.

All three jobs reduce to the same discipline, which is the discipline of reading a learned system as what it is: a statistical artifact whose behavior is a set of properties someone measured, on data someone chose, against thresholds someone set. Read it that way and the machine stops being either the terror of the op-ed or the miracle of the brochure. It becomes an auditable object, which is what every technology in this series becomes once its bones are showing.

One caution belongs in the front of the book rather than the back. The public conversation about military AI is dominated by two opposite errors: the apocalypse narrative, in which the machines are one training cycle from deciding to kill on their own, and the inevitability narrative, in which the machines are already so capable that a force which hesitates to delegate is a force that has already lost. Both narratives share a root: they treat the learning machine as a mind with intentions and a trajectory of its own. It is neither. It is an artifact with a distribution, fielded by an institution, inside a decision structure people designed, and every wonder and every atrocity it produces will be traceable to a threshold someone set, a dataset someone curated, a box someone drew, or a review someone waived.

The chapters that follow are an audit of those settings. The reader who finishes them will be able to look at any fielded system on any front page and answer the only question that matters about such a machine: what, exactly, was it decided to let it decide, and who signed for that decision?

A word on method before the chapters begin, because the series' method is a promise and the reader should know what was refused. No engagement described in this book is presented as history; the episodes are composites assembled from public reporting and named as composites. No vendor is named and no platform is endorsed, because platforms age on a budget cycle's schedule and the structures under them do not age at all: the confusion matrix, the confidence threshold, the data budget, the delegation box, and the audit trail are as true of next year's catalog as of this year's procurement record. Where the field is uncertain, the book says so and says why, and where the field's public conversation is wrong, the book disagrees in

the first person, because a guide that hedges its own judgments is a catalog wearing a coat.

The reader should also know the book's limits, stated up front rather than buried. This is not a legal text; the law of armed conflict appears where the engineering meets it, and the reader who needs the jurisprudence should read the jurists. It is not a primer on machine learning; the mathematics appears only in the plain arithmetic a decision-maker must command, because the reader who needs the derivations should read the mathematicians. And it is not the whole of the machine's story: the intelligence war, the industrial base, the nuclear shadow over any automated defense, all have their own volumes' worth, and this book touches each only where the military machine's anatomy requires it.

What it is, completely, is a commander's and a builder's guide to the learned weapon's structure, from the definition down, with the arithmetic shown, in the faith that has carried every volume of this series: that a reader who understands why a machine behaves as it does is prepared for the machine that has not been built yet.

The chapters, in the order the structure requires: the definition first, because a machine that cannot be classified cannot be governed; then the learning machine and its arithmetic, because everything it does is priced from that account; then perception, because the sensor pipeline is where the money and the mistakes concentrate; then the decision machinery, because delegation is a designed object; then the human, because the signatures must land somewhere; then the adversary, because the machine's brittleness is an attack surface; then the collective, because networks of machines have their own failure shapes; then verification, because a fielded system is a promise someone tested; then the record, where the claims are priced; and the decade ahead, in honest probabilities. Ten chapters, one structure, and one question asked of every machine the decade will field: who, exactly, can be asked about this decision, and what will they say?

Chapter 1: The Definition and Its Impostors

Every field begins with a definition problem, but military artificial intelligence begins with two, stacked: nobody agrees what artificial intelligence is, and nobody agrees what a military use of it is. The first disagreement is inherited from the parent discipline, where the label has been applied with equal enthusiasm to a chess program, a spam filter, a route planner, and a large language model, and where the definition on offer at any moment tends to track whatever the label's owner is trying to sell or to ban. The second disagreement is manufactured, deliberately, by institutions that profit from the blur: a vendor whose autopilot can be called autonomous buys a budget line; an advocacy organization whose target drone can be called a killer robot buys a headline; a program office whose classifier can be called a decision aid buys an exemption from the scrutiny a weapon deserves.

The blur is not a misunderstanding to be corrected with a better sentence. It is a terrain feature, and the first chapter of a book that means to audit the field has to map it.

The series method for cutting such a knot is the audit question: not what does the system do, but who can be asked about it. A decision procedure authored by a person can be traced to the person; a decision procedure induced from data can be traced only to the data and to whoever chose it. That distinction does the work a hundred definitions have failed to do, and it survives contact with marketing, because the impostors the field is full of all share one property: their procedures were written down before the mission, by named engineers, and can be inspected line by line.

Consider a torpedo from the middle of the last century. It homes on a sound or a wake; it arms at a set run distance; it turns on a set search pattern. Every one of those behaviors was specified, in a document, by people whose signatures are in the document's revision history, and any board of inquiry into the torpedo's behavior can read the specification and ask the signatories why line four reads as it does. The torpedo is automatic. It is not intelligent in the sense this book polices, and it is not the kind of machine the decade's laws and doctrines are being rewritten for, because its behavior is fully determined by its specification, and its failures are failures of the specification or of the hardware, which is precisely what makes it auditable.

Now consider a classifier trained to recognize vehicle convoys in overhead imagery. Its behavior is also fully determined, but not by a specification; it is determined by a training set, an architecture, and an optimization process whose result is a function nobody wrote. Ask the engineering team why the

classifier fires on this image and not that one, and the honest answer is a shrug rendered in mathematics: the difference lies somewhere in millions of parameters, induced from examples, and the best the team can do is measure the behavior on held-out data and report statistics.

Ask who decided the classifier should treat a bus depot as a convoy site, and the answer is not a name but a distribution: someone collected the examples, someone labeled them, and the labels taught the machine whatever the labelers believed. This is the grown rule, as opposed to the authored one, and the distinction is the book's working definition: a system is in this field's scope when its decision procedure was induced from data rather than authored by hand, and its governance follows from that fact rather than from the label on the brochure.

The definition has a useful corollary: automation and autonomy are different axes. Automation is about who executes a procedure; autonomy is about who selects among procedures, or sets their bounds. A thermostatically guided bomb is automated; a strike package whose targeting folders are updated aboard the aircraft by a system that scores candidates against a model is autonomous in the delegated sense, and the difference is not in the actuators but in the box. The book returns to this in Chapter 4 as the delegation box: the geometry, drawn by a person, inside which the machine is permitted to select.

The field's conversation is polluted by four recurring impostors, and each is worth naming because each has a constituency.

The first is the autopilot in fashionable clothing. A modern flight control system, a lost-link return route, a preplanned mission script with waypoints and contingencies: all are genuinely automated, genuinely useful, and genuinely not in this field's scope. The imposture begins when the label autonomous is applied to a script, because the word buys attention and budget. The audit question dismisses the claim in a minute: ask for the induced decision procedure, and the vendor produces a state machine, and the state machine has authors. The imposture is mostly harmless when a contractor commits it and mostly harmful when a treaty process commits it, because definitions written against scripts bind the wrong objects and leave the actual field ungoverned.

The second is the optimization in uniform. A route planner that solves a traveling-salesman problem over contested terrain, a scheduler that packs a strike package, a fuel allocator: all may use the same mathematics that powers learning systems, and all may be marketed as AI. But an optimizer that minimizes a written cost function is an authored artifact: the function has authors, the constraints have authors, and the board of inquiry can read both.

The learning in this book's scope enters only when the cost function itself, or the model it evaluates, was induced from data. The distinction matters because the two systems fail differently: the optimizer fails visibly, in a bad route someone can point at, while the learned system fails silently, in a confidence score someone trusted.

The third is the statistical alert dressed as a decision. A radar that reports a contact is a sensor; a fusion system that reports a track is arithmetic; a system that reports this track resembles a threat class learned from examples is in scope, and the imposture lies in pretending the resemblance report is an observation. It is not. An observation is what the antenna measured; a resemblance is what the model inferred; and the entire discipline of target review in the chapters that follow rests on keeping the two words apart, because the first can be recalibrated and the second can only be retrained.

The fourth impostor is the oldest: the magic word itself, applied to anything with a control loop. The word intelligence in artificial intelligence was a marketing choice made in the field's founding decade, and every generation since has inherited the bill. The machines this book describes do not understand the fight, do not intend anything, and do not know what a truck is; they compute functions over features, at scale, with a competence that is real and a comprehension that is zero, and the public conversation's two great errors, the terror and the miracle, both begin by granting the word its claim. The book will use the term learned system where precision matters and the label AI only where the field's institutions do.

The second half of the definition, the military in military AI, is less polluted but not empty. The audit question again does the work. A learned system is military not because it wears a uniform but because its employment structure is a command relationship: someone with authority ordered it into a decision chain, and its outputs have consequences the law of armed conflict reaches. The definition covers the obvious cases, the target recognizer and the sensor fusion engine and the electronic warfare cognitive system, and the less obvious ones: a predictive maintenance model that keeps a fleet flying is military AI in the logistic sense; a language model summarizing intercepted traffic is military AI in the intelligence sense; a system that scores candidates for a strike is military AI in the fullest sense, because the score enters a decision the law of war will later examine.

The book's center of gravity is the last class, the systems whose outputs feed lethal decisions, but the governing chapters apply to all three, because the audit discipline is identical even when the stakes are not.

A reader may ask why a book about engineering spends its first chapter on a

boundary. The answer is that every later failure mode in this book is a governance failure first, and governance fails at the boundary. A delegation policy that cannot distinguish a script from a learned system will either ban the script, which nobody needs, or excuse the learned system, which nobody should tolerate. A verification regime built on the specification will certify an authored system correctly and a learned system falsely, because the learned system has no specification to certify against.

A doctrine of human control written for the label rather than the mechanism will be satisfied by a signature block and empty of substance. The boundary is where the decade's real arguments, the treaty proposals, the export controls, the targeting reviews, will be won or lost, and the side that defines the objects precisely will write the rules the other side has to live inside. The chapters that follow therefore assume the working definition and use it as a blade: authored rules have authors; grown rules have data; and every claim that walks past wearing the label gets asked the same first question, which is the audit question, and every chapter ends by asking it of the machines that chapter described.

Definition: in scope is the system whose decision procedure was induced from data; out of scope is the authored script, the written optimizer, and the observation itself. Impostors: four, with constituencies, each dismissed by the audit question. The military half: command relationships, not camouflage. The stakes: every governance regime downstream of the definition fails at the boundary if the boundary is drawn wrong. The reader who has absorbed this chapter can already do what most of the field's public conversation cannot, which is to sort the front page: for any claimed autonomous weapon, ask whether its decision procedure was written or grown, and if written, read the specification and talk to its authors, and if grown, hold on, because everything in the next chapter is about what that means, and why the grown rule cannot be governed by the paperwork that governed the authored one.

The working definition earns its keep only when it is applied, so take three systems a reader will meet in the field's literature and run each through the audit question.

The first case is a wide-area surveillance aircraft with an onboard tracker. The vehicle orbits, the radar sweeps, and a processing chain converts reflections into tracks with headings, speeds, and histories, without which the raw returns would be an unreadable firehose. Is this artificial intelligence? Run the question: the tracker's association logic was authored, in a specification, decades mature, and its behavior on any given return can be traced to a rule with a paragraph number. It is automation, superbly engineered, and the

field's governance owes it the respect owed to any dangerous automation, review of the specification, testing of the edges, maintenance of the map data, and owes it nothing from the learned-system regime. The mistake to avoid is not classifying the tracker as AI; it is allowing a program office to attach the tracker's decades of certification to a learned component bolted beside it, because the certification of the authored half confers nothing on the grown half, and the integration is precisely where the laundering happens.

The second case is a target-recognition aid in a strike aircraft. The weapon systems officer sees candidate designations produced by a model trained on signature libraries, each with a score, and the officer's thumb hovers over an authorize switch. This is the field's center of mass, and the audit question splits it into two machines: the designation aid, grown, whose scores are statistics, and the authorize switch, authored, whose logic is the engagement sequence. The governance regime must treat the two differently, because the aid's failure is a distribution property and the switch's failure is a design property, and the officer's training must teach the difference, because an officer who believes the score is a measurement will treat the switch as a formality, and an officer who understands the score is an inference will read the switch as the moment the audit question gets answered by a human being. The case shows why the boundary matters operationally and not just jurisprudentially: the boundary runs through the cockpit.

The third case is a loitering munition with an onboard recognizer and a lost-link attack behavior. The brochure says autonomous. Run the question and the machine resolves into three parts: the flight control, authored and certified like the tracker's; the recognizer, grown, with all of Chapter 2's properties; and the delegation, the box of Chapter 4, which is the part that makes the machine a legal and moral object. The label autonomous describes the machine's public relations, but the machine's actual character is written in the box's four clauses, and two loitering munitions with identical brochures can be entirely different machines, one with a recognizer threshold set conservatively, a recourse clause that aborts on lost confidence, and a wall-review procedure in the maintaining unit, the other with a threshold set to keep the contractor's test accuracy, a lost-link attack as the recourse, and a wall nobody has read since the fielding brief.

The definition chapter's job is to make the reader see that the second machine exists behind the same label as the first, and that no vocabulary in the field distinguishes them, only the paperwork does.

The chapter has been hard on the blur, and honesty requires the charitable reading, because the blur has reasons. The first reason is that the boundary is

genuinely technical, drawn through the middle of systems that share hardware, contractors, and acronyms, and a legislator or editor cannot be expected to hold a confusion matrix in mind while drafting a resolution. The second reason is that the boundary moves with the tooling: a system whose rules were hand-authored last decade may be retrained this decade because the tooling made it cheaper, and a definition pinned to a technique ages faster than the fleet it governs, which is why the book's definition is pinned to the audit question rather than the method, because authored-versus-grown is a property of the evidence trail, not of the mathematics.

The third reason is that some blur is strategy: a state that has ratified nothing benefits from the ambiguity of the vocabulary, and a state that has fielded much benefits from the ambiguity of the label, and neither will thank the scholar who clarifies. The charitable reading grants all three reasons and still returns the same verdict, because the cost of the blur is paid downstream, by the review boards who certify against categories, the operators who train against labels, and the inquiries who arrive after the categories, labels, and reviews have all failed to name the thing that acted. Definitions are not decoration in this field. They are the first line of accountability, and every later chapter depends on the reader who refuses to let the boundary blur.

For the reader who wants the blade in one paragraph: a military system is in this book's scope when a material element of its deciding rule was grown from examples instead of being written by hand, such that its behavior cannot be fully reconstructed from any specification but only from the data, the architecture, and the training process, and such that the questions a board of inquiry will ask, why did it act, against what evidence, under whose authority, can be answered only by naming the dataset's custodians, the threshold's setters, and the box's authors, respectively. The three names in that sentence, custodian, setter, author, are the book's load-bearing roles, and the chapters that follow price every machine by what it demands of the three people who must be able to answer for it.

The boundary problem has a history, and the history explains why the blur persists, because each era's confusion was profitable to someone. The field's founding decade coined the label artificial intelligence as a fundraising term, and the bill has been arriving ever since: the label promised minds, the mathematics delivered functions, and the gap between the two has been rented out by every salesman, journalist, and legislator who needed a headline. The automation decades, from the torpedo through the cruise missile, produced genuine marvels of authored control, and the marvels inherited the label, so that a weapon that follows a terrain map written into its memory by engineers is called intelligent in the popular press, and the press's

usage then flows back into procurement documents, because a program office that must brief a committee borrows the committee's vocabulary.

The machine learning decades reversed the flow: systems that genuinely were grown, the recognizers and translators and fusion engines, were fielded under deliberately bland names, decision aids, cognitive assists, advanced processing, precisely because the bland names attracted less scrutiny than the label would have, and the field's governance energy spent itself on the frightening vocabulary while the actual statistical machinery entered the chains quietly. The history's lesson is the chapter's: the blur is not an accident of imprecision but an equilibrium of incentives, and the reader who wants clarity must expect to be fighting somebody's budget on every page.

For the reader who must brief others, the chapter's working definition collapses into a table that fits on one slide and answers the only question that matters at a review. Row one, the authored automaton: decision procedure written by hand, examples the torpedo, the waypoint script, the flight control; audit artifact, the specification; who answers, the spec's authors; regime, the century's test and evaluation. Row two, the authored optimizer: procedure written, the objective and constraints hand-designed, examples the route planner, the scheduler; audit artifact, the cost function; who answers, the function's authors; regime, the same, with the function's review added.

Row three, the grown component: procedure induced from data, inside a decision structure people designed, examples the recognizer, the classifier, the fusion aid; audit artifact, the dataset plus the clause file; who answers, the custodian, the setter, the author; regime, this book. Row four, the grown structure: multiple grown components coupled into a chain or a formation, examples the cued strike chain, the consensus swarm; audit artifact, the integration map plus the box ledger; who answers, the chain's owner; regime, this book, chapters 3, 4, and 7 especially. Four rows, three audit artifacts, one rule: the regime follows the evidence trail, never the label, and a reviewer who classifies every system by row before reading a page of its file has already done more governance than most of the field's published frameworks.

The definition will be stressed in ways the founding decades did not foresee, and the honest chapter names the stresses in advance. Foundation models, systems pretrained on corpora and adapted to many tasks, blur the row structure itself, because the training data is vast, largely civilian, and only partially known even to the trainer, and the adaptation data, the military examples, is the small governed part atop an ungoverned mass; the audit question survives, but the custodian's job changes, from owning a dataset to qualifying a foundation, and Chapter 2's provenance discipline scales to a new

order of difficulty.

Generative systems blur the evidence clause, because a system that composes text or imagery as part of a chain introduces outputs with no measurement behind them, Chapter 3's confabulated promoted from a failure mode to a feature. Learning at the edge, models that continue to adapt in the field, blur the wall review, because the model fielded is not the model flown a week later, and the ledger of Chapter 4 must grow a row for the model's own version history. Each stress has the same shape: the audit question still answers it, but the artifacts get heavier, and the field's institutions will be tempted, as always, to lighten the artifacts instead of building them, and a reader arriving here with the chapter absorbed will recognize the temptation in the vocabulary it arrives wearing, which is the vocabulary of agility, and the answer to it is the same answer the chapter gave the blur: the evidence trail is the machine, and no adjective lightens it.

The chapter's final instrument is the walkthrough the reader can run without a contractor in the room, because a definition is only as good as the drawer of files it opens. Take the force's own inventory, and for each system that anyone has ever called intelligent, autonomous, or cognitive, ask for three artifacts: the decision procedure's evidence trail, authored specification or grown dataset; the delegation's clause file, if the system selects anything; and the training corpus's coverage map, if the procedure is grown. The audit's outcome, from the composite experience of this book's method applied across services and decades, is a three-way sort.

The first pile is large: systems wearing the label that are authored automata, with excellent specifications, and the audit's finding is relief, plus the boundary discipline of not letting their certification launder the neighbors. The second pile is smaller and quieter: grown components fielded under bland names, decision aids and processing upgrades, and the audit's finding is the governance gap, because these entered service without the data regimes and clause files that the grown status requires, and the finding lands on whatever office owns fielding, which is where the decade's remediation work actually lives. The third pile is the smallest and most expensive: systems whose evidence trail cannot be produced, neither specification nor dataset, because the trail lives with a contractor, a predecessor program, or a merger, and the audit's finding is the field's hardest sentence: the machine is fielded and its behavior is unfalsifiable, which means the institution that owns it cannot answer for it, and the book's chapters that follow are the curriculum for the remediation, in the order the piles dictate: boundary first, data regime second, box third, human fourth, and the reader who has sorted the inventory honestly has already begun the governance, because the sort is the governance.

A last word to the reader who must brief the boundary upward, because the briefing is where definitions die or govern. Brief the audit question, not the taxonomy: in three sentences, tell the committee that some systems were written and can be inspected like bridges, that some were grown and can only be measured like weather, and that the field's entire governance problem is that both wear the same label and share the same brochures. Brief the three piles: written, governed by the old regime and fine; grown, governed by this book's regimes and mostly not yet; unprovable, a remediation list with names.

And brief the ask, which is small and decisive: every fielded system in the second and third piles gets sorted, filed, and signed, and the sort is one afternoon per system with the walkthrough of this chapter. The committee will not ask about philosophy, because the briefing has none; it will ask who does the work and what it costs, and the answers, a small office and less than the price of one bad inquiry, are the boundary's whole politics, and a reader arriving here with the chapter absorbed holds the only honest advantage the field's governance argument has ever had, which is that the work is genuinely cheap, and the only expensive part is deciding to start.

The chapter earns a closing anecdote, composited as the series requires, because definitions persuade better in story. An exercise, some years into the learning decade: two cells, side by side, both operating aids that score tracks for a strike cell. The first cell's aid was bought as a decision support modernization, and nobody in the cell had been told a model was in the loop; the scores arrived on the tracks like radar returns, and the cell treated them as sensor truth, and the exercise's injected errors sailed through, flagged with the machine's confidence, into two simulated engagements the after-action spent a week explaining.

The second cell's aid was the same model, same vendor, same scores, but the program had sorted it: grown, so it carried a coverage map, a label, and a brief, and the cell had been taught the machine's failure shapes on the drill floor. The injected errors were caught at a rate the after-action could publish. Same mathematics, same display, same war game, and the entire difference was the boundary work of this chapter: one institution knew what its machine was and governed it, the other did not and obeyed it. The anecdote is the book's whole argument in one afternoon, and the definition chapter ends where it began: the machines are different, the institutions are the same, and knowing which is which is the first weapon in the field, and it costs a briefing.

End of free sample. The full book is available at putrabooks.github.io